

Questions

1. .NET core is a free, open-source, cross platform development created by Microsoft for building Morden applications

It allows developers to create:

- Web applications
- Desktop applications
- Mobile apps
- Cloud services
- APIs
- Internet of Things (IoT) applications

Unlike the older .NET Framework, .NET Core runs on multiple operating systems such as:

- Windows
 - Linux
 - macOS
2. Key Benefits Include:
 - Cross platform support
 - High Performance
 - Open Source
 - Modern development tools
 - Security
 - Scalability
 - Cloud integration
 3. .NET supports multiple programming Languages, including:
 - C#
 - F#
 - Visual Basic(VB.NET)

Other languages can also work with .NET if they compile to intermediate Language (IL).

4. .Net Uses a common Language Runtime (CLR) and Common type System (CTS).

This means:

- Different Languages compile into the same Intermediate Language (IL).
- The CLR executes this shared Code.
- Developers can use multiple languages in one project.

Example:

- A class written in C@ can be used by a programming language written in VB.NE

This feature is called language interoperability

5. A Long-Term support (LTS) release is a version of .NET that receives
 - Security updates
 - Bug fixes
 - Stability improvements

For a longer support period (usually several years)

6. LTS releases are useful because they:
 - Provide stable and reliable environments
 - Reduce frequent upgrades
 - Are ideal for business and enterprise applications
 - Receive Long-Term security upgrades
 - Lower maintenance risk for production systems

Organizations prefer LTS versions when running important systems that must remain stable for a long time